Product Sales Enterprises Analysis

1. Project Overview and Objective

This project involves cleaning, transforming, and analyzing raw product sales and inventory data using Excel and Power Query, followed by structuring to derive meaningful business insights.

The main objective is to demonstrate data pre-processing techniques using Excel and including handling missing values, standardizing inconsistent formats, convert raw data into Pivot tables for summarization to enable accurate revenue and performance analysis across the online and retail sales based on product category wise and inventory-based analysis and an interactive Power BI dashboard visualization to make informative decisions.

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2. Data Sources

- **Source Description and Timeline:** Excelx.com/practice sales data and 2023-2025
- **Domain:** Online and Retail Sales Analytics.

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3. Problem Statement

1. Descriptive Analytics (What is happening?)

- **Identify top-performing product categories** (e.g., Electronics vs. Office Supplies) based on total transaction values
- **Rank retail stores and regions** (Store A, B, C, D) by revenue and item volume to pinpoint high-and-low performing locations
- **Regional manager efficiency** by analyzing total sales generated per staff member.

2. Diagnostic Analytics (Why is it happening?)

- **Referral Quality Analysis:** Correlate order cancellations and returns against traffic referral sources to determine if specific marketing channels attract low-intent buyers.
- **Customer Sentiment Diagnosis:** Cross-reference review ratings (e.g., 1-star ratings) with product categories to isolate specific quality or expectation issues driving customer rejection.

3. Predictive Analytics (What is likely to happen?)

- **Build a predictive model** using historical order data to forecast future purchase quantity based on the customer's initial referral source and demographic profile.
- **Predict peak purchasing timeframes** by evaluating historical time-of-day (Morning, Afternoon, Evening) and day-of-week trends to optimize staffing and digital ad scheduling.

4. Prescriptive Analytics (What should we do about it?)

- **Which suppliers** (Supplier) have the highest number of products currently marked as "Out of Stock" or "Low Stock"?
- What is the total Stock Value of items sitting in each Storage Location, grouped by their Stock Status?

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5. Tools & Technologies

- **Excel:** Data cleaning, Data transformation, And pivot tables.
- **Power BI:** Data modelling, DAX calculations, Visualization, and Interactive dashboard creation.

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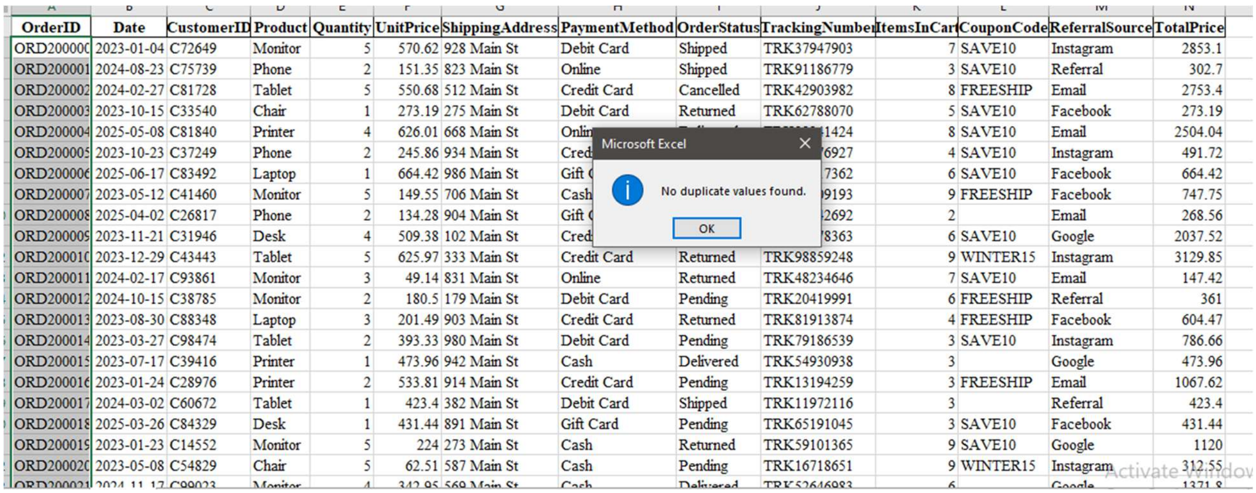
6. Data Pre-Processing:

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.

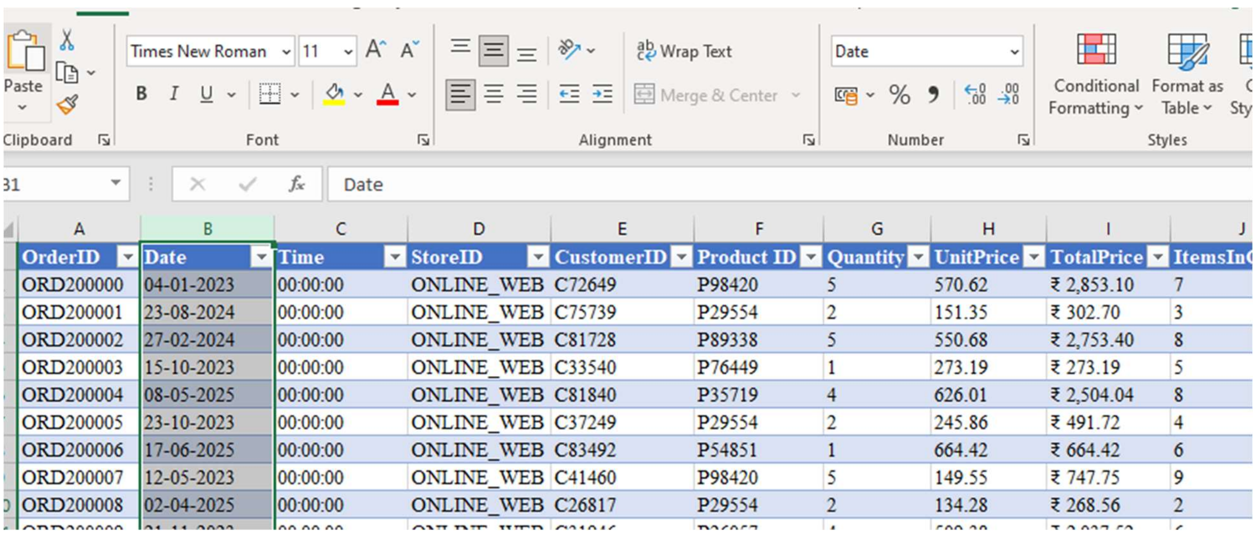
Removing Duplicates:



OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	CouponCode	ReferralSource	TotalPrice
ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FREESHIP	Email	2753.4
ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Onlin		1424	8	SAVE10	Email	2504.04
ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Cred		6927	4	SAVE10	Instagram	491.72
ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift		7362	6	SAVE10	Facebook	664.42
ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash		9193	9	FREESHIP	Facebook	747.75
ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift		2692	2		Email	268.56
ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Cred		8363	6	SAVE10	Google	2037.52
ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
ORD200012	2024-10-15	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending	TRK20419991	6	FREESHIP	Referral	361
ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FREESHIP	Facebook	604.47
ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3		Google	473.96
ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FREESHIP	Email	1067.62
ORD200017	2024-03-02	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped	TRK11972116	3		Referral	423.4
ORD200018	2025-03-26	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045	3	SAVE10	Facebook	431.44
ORD200019	2023-01-23	C14552	Monitor	5	224	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120
ORD200020	2023-05-08	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	312.55

Standardized Format:

Below showing as example, as formatted Date column as Date data type.



OrderID	Date	Time	StoreID	CustomerID	Product ID	Quantity	UnitPrice	TotalPrice	ItemsIn
ORD200000	04-01-2023	00:00:00	ONLINE_WEB	C72649	P98420	5	570.62	₹ 2,853.10	7
ORD200001	23-08-2024	00:00:00	ONLINE_WEB	C75739	P29554	2	151.35	₹ 302.70	3
ORD200002	27-02-2024	00:00:00	ONLINE_WEB	C81728	P89338	5	550.68	₹ 2,753.40	8
ORD200003	15-10-2023	00:00:00	ONLINE_WEB	C33540	P76449	1	273.19	₹ 273.19	5
ORD200004	08-05-2025	00:00:00	ONLINE_WEB	C81840	P35719	4	626.01	₹ 2,504.04	8
ORD200005	23-10-2023	00:00:00	ONLINE_WEB	C37249	P29554	2	245.86	₹ 491.72	4
ORD200006	17-06-2025	00:00:00	ONLINE_WEB	C83492	P54851	1	664.42	₹ 664.42	6
ORD200007	12-05-2023	00:00:00	ONLINE_WEB	C41460	P98420	5	149.55	₹ 747.75	9
ORD200008	02-04-2025	00:00:00	ONLINE_WEB	C26817	P29554	2	134.28	₹ 268.56	2

Conditional Formatting:

Showing the conditional formatting, In Total price column top 10 values are highlighted with pink color and filtered to find highest value of order.

OrderID	Date	Time	StoreID	CustomerID	Product ID	Quantity	UnitPrice	TotalPrice	D	Review Rating	Sales_
TX300222	26-06-2025	09:52:00	S4	Retail-customer	P29554	10	399.74	₹ 3,997.40	0	0	Retail
TX300339	28-11-2023	19:20:00	S9	Retail-customer	P76449	10	392.71	₹ 3,927.10	0	0	Retail
TX300463	24-04-2025	10:55:00	S3	Retail-customer	P26057	10	394.91	₹ 3,949.10	0	0	Retail
TX300489	27-12-2023	11:12:00	S8	Retail-customer	P76449	10	391.43	₹ 3,914.30	0	0	Retail
TX300732	15-02-2023	15:10:00	S10	Retail-customer	P89338	10	390.62	₹ 3,906.20	0	0	Retail
TX301019	20-04-2023	19:11:00	S10	Retail-customer	P35719	10	396.16	₹ 3,961.60	0	0	Retail
TX301108	24-03-2023	12:28:00	S10	Retail-customer	P35719	10	386.21	₹ 3,862.10	0	0	Retail
TX301146	20-09-2024	10:44:00	S8	Retail-customer	P76449	10	393.44	₹ 3,934.40	0	0	Retail
TX301840	01-12-2024	15:32:00	S6	Retail-customer	P98420	10	388.80	₹ 3,888.00	0	0	Retail
TX301954	17-10-2024	17:54:00	S3	Retail-customer	P54851	10	391.42	₹ 3,914.20	0	0	Retail

Creating new column for analysis requirement:

- **Create a Product Category column from Customer purchase history**

  =XLOOKUP([@Product],Customer_Purchase_History[Product],Customer_Purchase_History[ProductCategory],"Not Found")

- **Creating stock status column**

  =IF([@QuantityInStock]<=[@ReorderPoint]," ⚠ Reorder Now ", " ✅ Stock OK")

- **Creating Sales_channel column**

  =IF(LEFT([@OrderID],1) = "O", "Online", "Retail")

Convert the data into Fact and Dimension Table:

The data has two different sales channels online, Retail sales order data. So, Convert the data into Fact and Dimension Table.

Fact Sales Table and Dimension Table		
Fact _Sales Column Name	**Dimension Table Column Name**	**Related Column**
Fact_Sales	Dim_Product Details	Product Id
Fact_Sales	Dim_Order Details	Order Id
Fact_Sales	Dim_Location	Store Id
Fact_Sales	Dim_Time	Order Id

7. Data Modelling and DAX (Power BI)

Data Model:

- **Fact sales table into Dim_Product Details:**

  The Fact_sales table connected by the product id of the Dim_Product details to view the category wise revenue & get know the overall analysis by product category wise.

  Type: M:1 **(**Many to One)

- **Fact sales table into Dim_Location Table:**

  The Fact_sales table connected by the Store id of the Dim_Location table to view store wise sales performance and regional manager performance by revenue.

  Type: M:1 (Many to One)

- **Fact sales table into Dim_sales order details:**

  The Fact_sales table connected by the order Id of Dim_sales order details table to get know the based on the referral source how many orders, and which area we are focusing into improve our marketing

  Based on the Referral source & Payment method we can ably predict future orders.

  Type: 1:1(One to One)

- **Fact sales table into Dim_Time Details:**

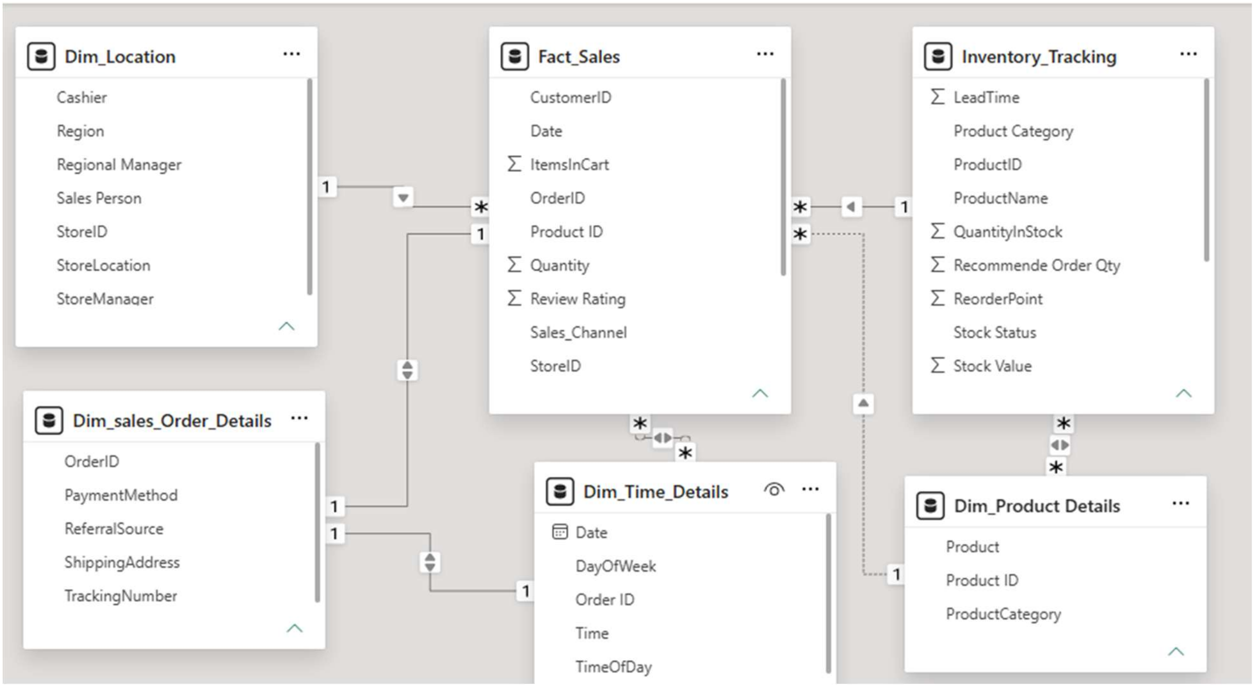
  The Fact_sales table connected by the date of Dim_time details table to get know which day and what time the order is high and low and predict future model.

- **Fact_sales table into Inventory Tracking:**

  The Fact_sales table connected by the product id of Inventory_Tracking table to visualize the inventory analysis in the dashboard by product category wise.

- **Dim_Time into Dim_sales order details:**

  The Dim_sales order details table connected by order id of Dim_Time table to forecast future order quantity based on the customer's initial referral source and demographic profile.



Calculated Columns & DAX Measures:

Dax Measure	Attribute	Used
Rev Per store	Store ID, Total Revenue	To analyze regional Manager Performance
Total Stock Value	Quantity in stock, Unit Price	To analyze stock status & stock value
Order count	Order Id	To analyze the Referral quality, Review rating
Low stock count	stock status	To analyze which supplier has lowest number of stock

Formula Used:

- Rev Per Store = DIVIDE ([Regional Revenue], DISTINCTCOUNT (Dim_Location [StoreID]), 0)
- Total Stock Value = SUM (Inventory_Tracking [Stock Value])
- Order count = DISTINCTCOUNT (Fact_Sales [OrderID])
- Low Stock Count = CALCULATE(COUNTROWS(Inventory_Tracking)

8. Analysis and Visualizations (Power BI):

Visualizations:

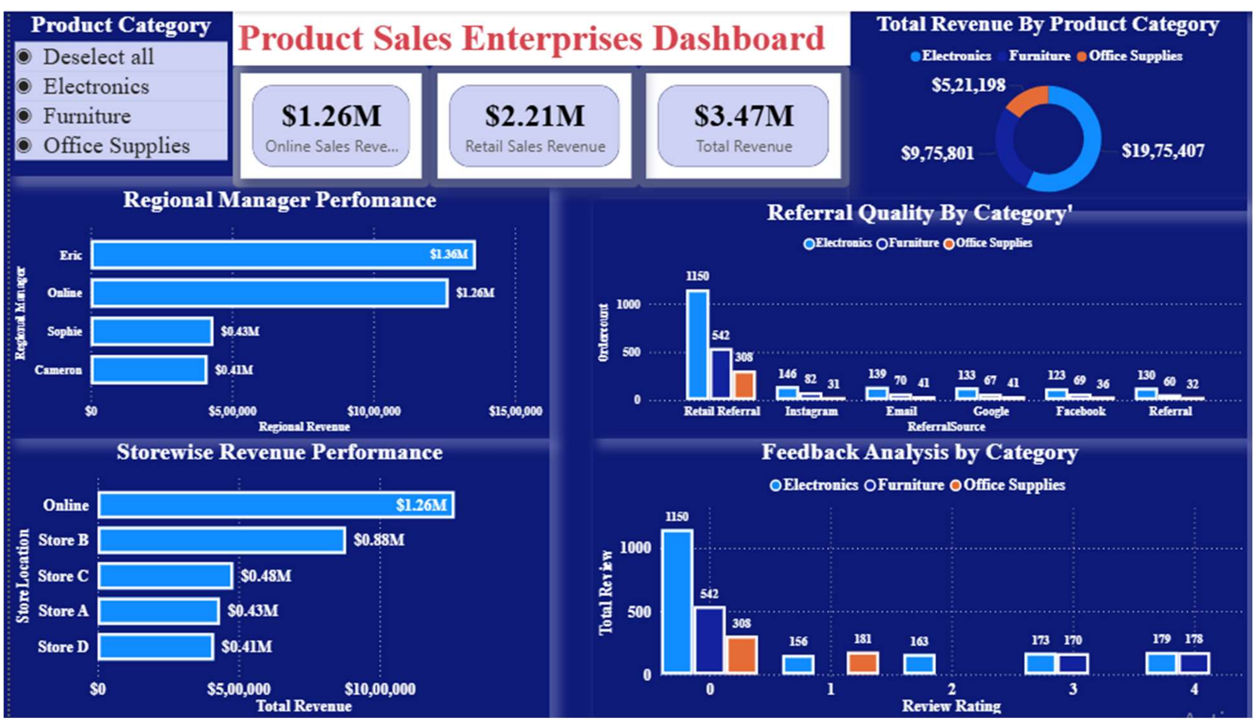
Dashboard 1:

- **KPI Card:** Total Revenue, Online sales Revenue, Retail sales Revenue
- **Donut chart:** Product category wise sales revenue.
- **Cluster Bar Chart:** Store wise sales performance, Regional Manager performance by revenue.
- **Cluster Column chart:** Referral quality analysis, Feedback analysis.
- **Slicer:** Showing Product category

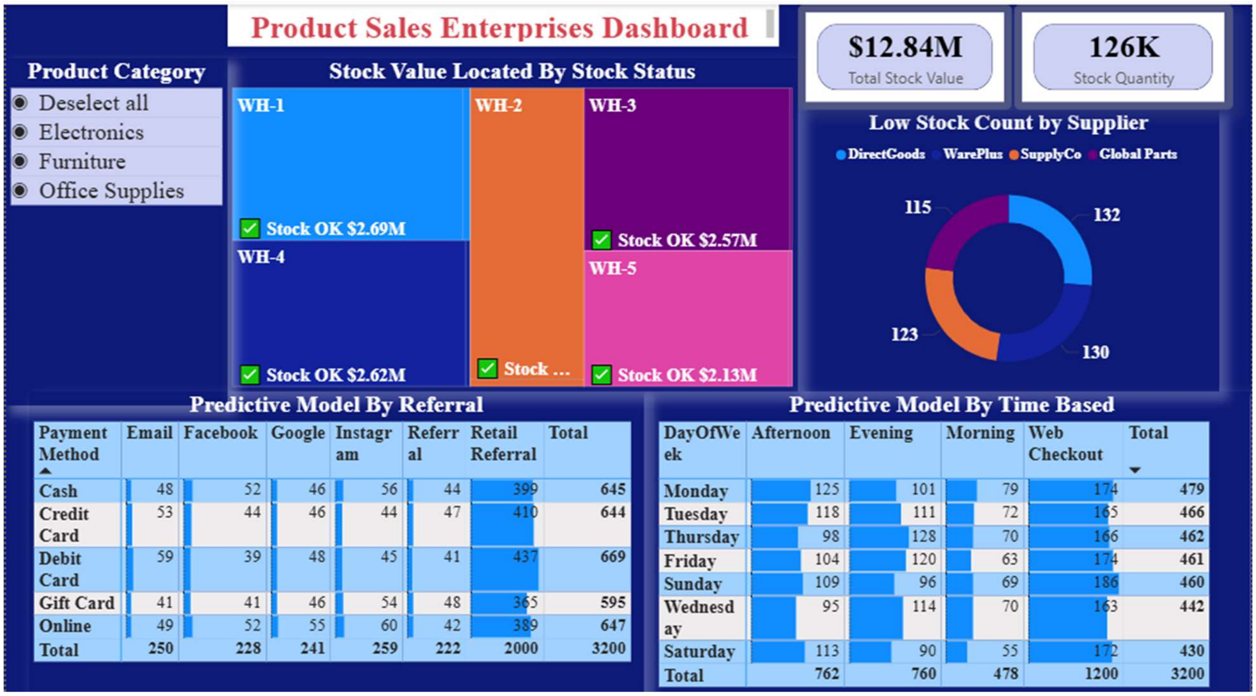
Dashboard 2:

- **KPI Card:** Total Stock value. Total stock quantity.
- **Donut Chart:** Low stock count by supplier
- **Tree map:** Stock value located by stock status
- **Matrix:** Predict model by referral, predict model by time based
- **Slicer:** Showing Product category.

Consolidate Dashboard 1:



Consolidate Dashboard 2:



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9. Insights & Conclusions:

Key findings:

- Total revenue in both sale channel $ 3.47 M around the 3200-order count.
- Retains the total revenue by online sales revenue $1.26 M and retail sales revenue $2.27M
- Product categorical wise Electronics items are leading revenue $19.7M followed by Furniture $ 9.7M and Office supplies $ 5.2M.
- Exploring region wise and regional manager efficiency the highest revenue in East region by Eric Regional Manager $1.36M followed by south region by Sophie $ 0.43M and Central region Cameron $0.41M
- Store wise revenue Store B leasing highest revenue $ 0.88 M followed by Store c $ 0.48M, Store A $ 0.43M and next Store D $ 0.41M.
- Online sales revenue remains lower than retail revenue. To identify our performance gaps, we are analyzing order counts across different referral sources and review rating.

- Build a predictive model by online sales and Instagram slightly leads online order channels with 259 orders, while standard Referrals trail at 222.
- Customers consistently prefer Debit Cards (669 total orders) over Gift Cards (595 total orders) across all channels.
- Web Checkout referred as online sales and dominates digital transactions with 1,200 orders, vastly outperforming specific daily time slots.
- On Retail Sales - Mondays and Tuesdays peak as the highest-volume shopping days, while Saturday drops to the weekly low.
- Total stock value sits healthy at $12.84M across 126K items, split stably between major warehouses.
- Low stock risks are balanced evenly among four main suppliers, reducing single-supplier reliance for critical items.

Analysis Report:

- Total revenue $ 3.47 M across the online ($ 1.26M) and retail ($ 2.26M) sales channel.
- Identifying top performing product category base on the total revenue.
- Rank regions and stores (A through D) by revenue and volume to identify high and low performers.
- Evaluate regional manager efficiency by calculating total revenue.
- Analyze referral quality by linking total order counts to marketing channels to identify top-performing sources.
- Evaluate customer experience by linking reviews to specific product categories to isolate quality issues.
- Forecast future purchase totals by training a predictive model on customer demographics and initial referral sources.
- Forecast peak purchasing windows by analyzing historical time and day trends to optimize staffing and ad schedules.
- Identify which suppliers have the highest number of out-of-stock or low-stock products.
- Calculated total stock value for each warehouse, grouped by stock status of low stock or Stock ok.

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10. Conclusions:

The excel and power BI integration proved effective for end-to-end data analysis, from raw data to visual reporting. Using Excel for cleaning and transforming the data and Power BI for modeling. Key tasks included establishing a star schema, creating calculated fields, and applying DAX measures for descriptive, diagnostic, predictive, and prescriptive analysis.

Thank You!!!!